Software Documentation

1. Software Overview

- *What does the software do?*
- *Platforms used: MicroPython, MIT App Inventor*

The app does not have much purpose on the functional end, apart from connecting to the esp32 via bluetooth, marking the start of the timer which controls the LED's inception and the beginning of the tally recording. The app merely only interacts with the user and aids in creating a lighthearted atmosphere - the most important one playingthe audio itself. Micropython is the logical centre of the system. It keeps a track of user tallies, compares it against the control dataset, and calculates the final accuracy score of the user. Micropython also controls the pattern of the LED.

2. System Logic

Explain flow in simple steps

1. *User connects to bluetooth app*
2. *User presses start button and places one hand over the physical button of the stage*
3. *When the song begins, there is a time window at every 'oiia' where the code will add to a tally if the button is clicked accurately.*
4. *Along with a tally, the amount of time the button has been pressed down for will also be recorded.*
5. *The "pressed-down" values are compared against a control dataset, and each tally is assigned an accuracy value, as determined by the difference of the control and user input*
6. *At the end of the song, the cumulative accuracy value score will appear.*
7. *The score is only calculated when the number of user input tallies is equal to the control tallies.*

3. Flowchart

https://miro.com/app/board/uXjVHbwkr9g=/

4. App Flow (If MIT App Inventor)

https://miro.com/app/board/uXjVHbwkr9g=/